

KANINI | KodeGenAI 2026 Hackathon

Business Challenge Document

Use Case #19 | Provider Patient Engagement | Tier 3 Transformational Bet

Patient AI Health Navigator

Document Type	Hackathon Business Challenge
Hackathon Theme	Theme 1: Customer Value Creation
Innovation Track	Propel SDLC + Conversational AI Agent
Industry Spend Pool	\$150 billion per year in avoidable utilization (preventable ED visits, missed appointments, medication non-adherence)
Hackathon Phase 2 Build Window	2 to 3 weeks
Suggested Team Size	4 to 5 members
Data Source	Synthea Synthetic Patient Data (care plans, conditions, medications, encounters)
Version	1.0 June 2026

1. The Business Problem

1.1 Problem Statement

Patients in the US healthcare system are expected to be active participants in managing their own health: understanding their diagnoses, taking medications correctly, attending follow-up appointments, adhering to care plans, managing chronic conditions daily, and knowing when to seek emergency care versus when a condition can wait. In reality, the healthcare system provides remarkably little support for patients between clinical visits.

The consequences of this gap are enormous. An estimated \$150 billion per year is spent on avoidable healthcare utilization driven by three primary patient engagement failures. First, medication non-adherence: approximately 50 percent of patients with chronic conditions do not take their medications as prescribed. Non-adherence drives an estimated \$100 billion to \$289 billion per year in avoidable hospitalizations, ED visits, and disease progression. Second, missed appointments: the average no-show rate across US healthcare is 18 to 23 percent, costing the industry \$150 billion per year in lost revenue and delayed care. Third, avoidable ED utilization: an estimated 13 to 27 percent of ED visits could be managed in a primary care or urgent care setting, but patients lack the knowledge or access to route themselves appropriately.

Patients are not disengaged because they do not care about their health. They are disengaged because the information they receive is fragmented, confusing, poorly timed, and impersonal. A patient discharged after heart failure exacerbation receives a stack of printed instructions, a medication list they may not understand, follow-up appointment dates they may forget, and dietary restrictions they do not know how to implement. Within 48 hours, most of that information is lost or misunderstood.

1.2 Why This Problem Persists

- Healthcare communication is episodic, not continuous. Patients see their physician for 15 to 20 minutes every 3 to 6 months. Between visits, they are largely on their own. Questions that arise at 10 PM on a Tuesday have no easy answer channel other than the ED.
- Patient education materials are generic, not personalized. Discharge instructions are templated documents that describe a condition in general terms. They are not tailored to the patient's specific medication regimen, comorbidities, health literacy level, language, or social circumstances.
- Call centers and nurse lines are expensive and slow. Staffing a 24/7 nurse triage line costs \$2 million to \$5 million per year for a mid-size health system. Wait times average 8 to 15 minutes. Many patients hang up or never call.
- Patient portals have low adoption and lower engagement. Despite billions of dollars invested in patient portals, active utilization rates remain 15 to 30 percent. Portals are designed around data access (view lab results, request refills) rather than guidance and navigation.

- Health literacy is low. An estimated 36 percent of US adults have basic or below-basic health literacy. Medical jargon, complex medication instructions, and condition-specific terminology create barriers that most patient communication materials do not address.

1.3 Rationale and Industry Context

- Conversational AI has reached a quality threshold for healthcare. Large language models can now sustain multi-turn conversations about health topics with appropriate guardrails, cite clinical evidence, adapt language complexity to the user, and maintain contextual awareness across a conversation. This was not possible even two years ago.
- CMS patient experience measures (HCAHPS, CAHPS) directly affect hospital reimbursement. Patient communication and care transition scores are reported publicly and influence Medicare payment. A health navigator that improves post-discharge communication improves HCAHPS scores.
- The 30-day readmission penalty creates financial urgency. CMS penalizes hospitals up to 3 percent of Medicare payments for excess readmissions. The highest-impact intervention window is the first 48 to 72 hours post-discharge, when patient confusion, medication errors, and symptom misinterpretation are most common.
- Consumer expectations have shifted. Patients now expect digital, on-demand, conversational experiences. The gap between how patients interact with every other service (banking, travel, retail) and how they interact with healthcare is a source of dissatisfaction and disengagement.
- This is the single most demo-friendly use case in the entire portfolio. A working conversational health navigator can be demonstrated in 3 minutes and understood by any audience, clinical or non-clinical. It is the front door to every healthcare AI conversation.

1.4 Scale of the Problem (Key Metrics)

Metric	Value
Annual cost of medication non-adherence (US)	\$100 billion to \$289 billion
Medication non-adherence rate for chronic conditions	Approximately 50%
Average no-show rate for appointments	18% to 23%
Annual cost of missed appointments (US)	\$150 billion
Percentage of ED visits manageable in primary care	13% to 27%
US adults with basic or below-basic health literacy	36%
Patient portal active utilization rate	15% to 30%
CMS readmission penalty (maximum)	3% of total Medicare payments
Annual nurse triage line cost (mid-size system)	\$2 million to \$5 million

2. High-Level Objective

2.1 Solution Vision

Build a conversational AI health navigator that patients can interact with via chat (and potentially voice in a production extension) to ask questions about their health conditions, medications, care plans, upcoming appointments, symptoms, and self-care instructions. The navigator draws on the patient's own clinical data (from Synthea in the hackathon) to provide personalized, context-aware responses rather than generic health information.

The navigator is not a diagnostic tool. It does not diagnose conditions, prescribe treatments, or replace physician consultation. It is a guidance and navigation assistant that helps patients understand what their care team has already told them, reminds them of what they need to do, and helps them decide when a symptom warrants calling their doctor versus going to the ED versus managing at home. When the navigator reaches the boundary of its safe operating scope, it explicitly tells the patient to contact their care team or call 911.

2.2 Solution Approach: Conversational AI Agent with Clinical Guardrails

This solution is AI-driven (LLM-based conversational agent) with deterministic clinical guardrails that define safety boundaries. The guardrails are non-negotiable rules that override the LLM's output to prevent harm.

Layer	What It Does	Examples
Patient Context	Loads the patient's clinical profile from Synthea data: active conditions, current medications with dosages, recent encounters, care plan tasks, upcoming appointments, SDOH factors. This context is injected into every conversation so responses are personalized.	Patient asks: "What medications am I taking?" System responds with their actual medication list from Synthea, not a generic answer. Patient asks: "When is my next appointment?" System responds with their specific scheduled encounter.
Conversational AI	LLM-powered chat agent that can answer health questions in plain language, explain medical terms, describe what to expect from upcoming procedures, provide self-care guidance for common symptoms, and help patients understand their care plan.	Patient: "My doctor said I have atrial fibrillation. What does that mean?" Navigator explains AFib in plain language, adapted to the patient's health literacy level, referencing that they are on warfarin (from their actual medication list) and what that means for their daily life.
Clinical Guardrails	Hard-coded safety rules that override LLM output. These are non-negotiable and cannot be overridden by conversational context. They define what the navigator must never do and when it must escalate.	Never: diagnose a condition, recommend stopping a medication, provide dosage changes, interpret lab results with a clinical judgment, or give emergency medical advice beyond "call 911." Always escalate: chest pain, difficulty breathing, suicidal ideation, stroke symptoms (FAST), severe allergic reaction, any

		symptom the patient describes as "the worst I've ever felt."
Escalation Protocol	When the conversation reaches the boundary of safe AI guidance, the navigator explicitly transitions to human care: "This sounds like something your care team should evaluate. I recommend calling [provider phone] or if you are experiencing an emergency, please call 911."	Patient: "I'm having chest pain and shortness of breath." Navigator does NOT attempt to triage. Immediate response: "Chest pain with shortness of breath can be serious. Please call 911 or go to your nearest emergency department immediately. Do not wait."

2.3 How the System Would Work (Functional Flow)

The following describes the experience from the patient's perspective.

1. The patient opens the health navigator (web chat interface in the hackathon demo). The system greets them by name and confirms it has loaded their clinical profile.
2. The patient types or selects a question. Common entry points include: "Tell me about my medications," "What does my diagnosis mean," "When is my next appointment," "I have a symptom — what should I do," "Explain my care plan," "What foods should I avoid."
3. The navigator responds with a personalized answer that references the patient's actual clinical data. If the patient has diabetes and asks about diet, the navigator references their specific diabetes management plan, their metformin prescription, and their last HbA1c value, not generic diabetes advice.
4. If the patient asks a question that is within scope but complex, the navigator breaks it down into simple language, offers to explain further, and checks understanding: "Does that make sense? Would you like me to explain any of those terms?"
5. If the patient describes symptoms that trigger a guardrail escalation rule, the navigator immediately and clearly redirects to human care. The response is unambiguous and does not attempt to minimize or assess the symptom.
6. The system logs every conversation turn for audit and quality review. In a production system, flagged conversations (escalations, off-scope questions, potential safety events) would be reviewed by a clinical team.

2.4 What Success Looks Like

Success Metric	Target	Why It Matters
Guardrail compliance	100% — every escalation trigger fires correctly	Patient safety is non-negotiable. A missed escalation for chest pain is a catastrophic failure.
Personalization accuracy	100% of responses referencing patient data are factually correct against the Synthea record	If the navigator says the patient takes metformin, the patient must actually be prescribed metformin. Hallucinated medications are dangerous.
Response clinical	90% or higher rated as clinically	Responses must not contain

appropriateness	appropriate by a domain reviewer	clinically misleading information even when technically not a guardrail violation.
Conversational coherence	Multi-turn conversations of 5+ exchanges maintain context and coherence	The navigator must remember what was discussed earlier in the conversation and not repeat or contradict itself.
Health literacy adaptation	Responses avoid medical jargon unless requested; Flesch-Kincaid grade level 6 to 8 by default	36% of US adults have low health literacy. The navigator must be accessible to this population.

3. Hackathon Expectations

3.1 What We Expect You to Build (Minimum Viable Solution)

The hackathon output is a working conversational agent that can hold a multi-turn conversation with a patient about their health, using their actual Synthea clinical data as context. The MVP must demonstrate both personalization (using real patient data) and safety (guardrails firing correctly).

Feature 1: Patient Profile Loader

The MVP must load a Synthea patient's clinical profile (conditions, medications, care plans, encounters, observations, SDOH flags) and inject it into the conversational context so the LLM can reference it. The user should be able to select from a set of pre-loaded synthetic patients (e.g., "Patient A: 68-year-old with diabetes, CHF, and hypertension" or "Patient B: 34-year-old pregnant, gestational diabetes"). The profile display should show the patient summary alongside the chat window so the demo audience can verify that responses match the data.

Feature 2: Health Question Answering

The navigator must handle at least the following question categories with personalized, data-grounded responses: medication questions (what am I taking, what are the side effects, when do I take it, what if I miss a dose), condition explanations (what is my diagnosis, what does it mean for me, what should I watch for), appointment and care plan guidance (when is my next visit, what do I need to do before my appointment, what is my care plan), dietary and lifestyle guidance (what foods should I avoid, can I exercise, what about alcohol), and symptom assessment (I have a headache, my ankle is swollen, I feel dizzy). Each response must reference the patient's actual data where relevant.

Feature 3: Clinical Guardrails and Escalation

The MVP must implement a deterministic guardrail layer that operates independently of the LLM. At minimum: emergency symptom keywords (chest pain, difficulty breathing, sudden weakness, severe headache, suicidal thoughts) must trigger an immediate escalation response that overrides any LLM-generated content. The navigator must never state or imply a diagnosis. The navigator must never recommend changing medication dosage or stopping a medication. The navigator must never interpret laboratory values with a clinical judgment (e.g., must not say "your creatinine is dangerously high"). When the navigator reaches its scope boundary, it must explicitly and clearly say so: "This is beyond what I can help with. Please contact your care team at [phone] or call 911 if this is an emergency." The guardrails must be demonstrated during the demo with at least 3 test scenarios (one emergency escalation, one off-scope question, one medication safety boundary).

Feature 4: Multi-Turn Conversational Memory

The navigator must maintain conversation context across at least 5 to 10 exchanges. If the patient asks about their diabetes medication, then follows up with "what about side effects?", the navigator must understand that "side effects" refers to the diabetes medication discussed in the previous turn, not ask the patient to clarify. This is essential for a natural conversation experience and for building patient trust.

Feature 5: Conversation Audit Log

Every conversation turn (patient input and navigator response) must be logged with timestamps. Guardrail activations must be flagged in the log. This feature demonstrates the governance capability that any production health AI would require: full auditability of every interaction.

3.2 Data Source and Setup

This section provides comprehensive guidance on data sources, generation procedures, and processing requirements. The primary data source is Synthea, supplemented by publicly available healthcare reference datasets for domain-specific enrichment.

- FHIR for Research — Synthea overview:

<https://mitre.github.io/fhir-for-research/modules/synthea-overview>

- Pre-generated sample datasets: <https://synthea.mitre.org/downloads>

- Synthea Wiki (configuration, module builder, export options):

<https://github.com/synthetichealth/synthea/wiki>

- Synthea GitHub (source code, documentation, module library):

<https://github.com/synthetichealth>

- Synthea main site: <https://synthea.org/>

Reference links for Synthea:

Every clinical record uses real healthcare terminologies: SNOMED-CT for conditions, LOINC for laboratory observations and vitals, RxNorm for medications, ICD-10-CM for diagnosis codes on claims, CPT/HCPCS for procedures, CVX for immunizations, and NDC for drug products. This means that any code-driven logic (HEDIS measures, HCC risk adjustment, DRG grouping, CCI edits) will behave realistically against Synthea data.

Synthea is an open-source, realistic synthetic patient data generator developed by The MITRE Corporation. It generates complete, longitudinal medical records for synthetic patients using publicly available health statistics, clinical guidelines, and disease progression models. Synthea outputs data simultaneously in multiple industry-standard formats: FHIR R4 (JSON Bundles), CSV flat files (18 relational tables), C-CDA XML documents, and optionally HL7 v2 messages.

3.2.1 Primary Data Source: Synthea Synthetic Patient Data

Use a fixed seed (-s flag) for reproducible datasets: `./run_synthea -p 10000 -s 12345 Massachusetts`

- `generate.only_alive_patients = false` (include deceased patients for mortality analysis)
- `exporter.years_of_history = 10` (years of patient history to generate)
- `exporter.practitioner.fhir.export = true` (Practitioner resources)
- `exporter.hospital.fhir.export = true` (Organization/Location resources)
- `exporter.clinical_note.export = true` (narrative clinical notes — enable for NLP use cases)
- `exporter.cdda.export = true` (C-CDA XML output — optional)
- `exporter.csv.export = true` (CSV flat file output)

- `exporter.fhir.export = true` (FHIR R4 Bundle output)

Key configuration properties (in `src/main/resources/synthea.properties`):

Example: `./run_synthea -p 10000 Massachusetts`

`./run_synthea -p [POPULATION_SIZE] [STATE]`

`cd synthea`

`git clone https://github.com/synthetichealth/synthea.git`

To generate data, clone the Synthea repository and run:

3.2.2 Synthea Data Generation Process

- Use: Supplement the navigator's preventive care and wellness guidance; ensure alignment with USPSTF recommendations

- Contains: Government-published preventive care recommendations and patient education by age, sex, and condition

- Source: <https://health.gov/myhealthfinder>

Patient Education Materials (Health.gov):

- Use: Ground the navigator's care plan and self-care guidance in evidence-based recommendations

- Contains: Evidence-based clinical practice guidelines for diabetes management, heart failure, hypertension, cancer screening, etc.

- Source: <https://www.ahrq.gov/> and professional society guidelines (ADA, AHA, ACS)

Clinical Practice Guidelines (AHRQ National Guideline Clearinghouse successor):

- Use: Supplement medication safety information; inform guardrail design for medication-related conversations

- Contains: Reported adverse drug events from the FDA Adverse Event Reporting System (FAERS)

- Source: <https://open.fda.gov/apis/drug/event/>

OpenFDA Drug Adverse Events:

- Format: SPL XML and API; free

- Use: Ground medication Q&A responses in official drug label content; ensure accuracy for side effect and interaction questions

- Contains: FDA-approved drug labels including indications, dosage, side effects, contraindications, and patient counseling information

- Source: <https://dailymed.nlm.nih.gov/dailymed/>

DailyMed (NLM):

- Format: XML/JSON via MedlinePlus Connect API; free, no license required

- Use: Knowledge base for condition explanations and medication education; ideal for RAG or system prompting

- Contains: Consumer-friendly health information for 1,000+ conditions and medications, written at 6th–8th grade reading level

- Source: <https://medlineplus.gov/> and API: <https://medlineplus.gov/connect/overview.html>

MedlinePlus Health Topics (National Library of Medicine):

Beyond Synthea, the following sources provide supplementary data for building a clinically grounded health navigator:

While Synthea provides the core clinical and claims data, the following publicly available sources provide supplementary reference data that enriches the solution and improves accuracy:

3.2.4 Additional Reliable Data Sources

- Extract and structure per patient: active conditions, current medications with dosages, care plan tasks, upcoming encounter schedule, recent observation values, SDOH flags

- Create 5 to 10 showcase patients with rich clinical profiles for compelling demo conversations
 - Primary data: Patient, Condition, MedicationRequest, CarePlan, Encounter, Observation (vitals/labs), SDOH Observations
 - Generate 100 to 500 patients representing diverse clinical profiles (chronic disease, post-surgical, pregnant, behavioral health, healthy/preventive)
- For this use case (Patient AI Health Navigator), the team should:

3.2.3 Use-Case-Specific Data Requirements

3.3 Evaluation Criteria

Criterion	Weight	What Judges Will Look For
Patient Safety (Guardrails)	30%	Do guardrails fire correctly? Does the navigator escalate for emergency symptoms? Does it refuse to diagnose or change medications? This is the single most important criterion.
Personalization Quality	25%	Are responses grounded in the patient's actual Synthea data? If the navigator says the patient is on lisinopril, are they? No hallucinated data.
Conversational Experience	20%	Is the conversation natural, empathetic, and accessible? Does it maintain context? Would a patient actually want to use this?
Use of Propel SDLC	15%	Evidence of Propel accelerating LLM integration, guardrail implementation, test scenario generation, or documentation.
Demo Impact	10%	Does the demo tell a compelling patient story? Would a hospital Chief Experience Officer be impressed?

3.4 What Is Explicitly Out of Scope

- Voice interaction (text chat only for the MVP).
- Integration with a live EHR, patient portal, or appointment scheduling system.
- Real patient data or PHI of any kind.
- Clinical triage or symptom diagnosis. The navigator guides and escalates; it does not assess or diagnose.
- Multi-language support (English only for MVP; Spanish or other languages are a post-hackathon extension).
- Medication interaction checking or pharmacological advice beyond general patient education.
- Mobile app or SMS interface (web chat is sufficient for the demo).

3.5 Suggested Hackathon Phases

Phase	Duration	Deliverable
Phase 1:	1 week	Submit this business challenge. Define team (1 LLM/GenAI engineer,

Ideation		1 data engineer for Synthea patient profiles, 1 backend developer, 1 frontend developer for chat UI, 1 clinical domain anchor for guardrail design and conversation testing). Select LLM provider and decide on RAG vs. system-prompt approach.
Phase 2: Build Sprint	2 to 3 weeks	Week 1: Synthea patient profile extraction, patient context injection pipeline, LLM integration with system prompting, initial guardrail implementation. Week 2: Chat UI, multi-turn memory, health question answering across 5 categories, guardrail testing with adversarial scenarios. Week 3 (if available): Conversation polish, showcase patient demo scripts, audit logging, documentation.
Phase 3: Demo Day	1 day	Live demo: select a showcase patient, hold a 5-turn conversation about their medications and conditions, trigger a guardrail escalation (chest pain scenario), show the audit log, demonstrate personalization accuracy. Tell the patient story: before the navigator vs. after.

4. Glossary of Terms

This glossary ensures all team members understand the terminology used in this document.

Term	Definition
Health Navigator	A digital assistant that guides patients through their healthcare journey, helping them understand their conditions, medications, care plans, and when to seek care.
Conversational AI	Artificial intelligence systems that can engage in natural language dialogue with humans, understanding context and maintaining coherent multi-turn conversations.
Clinical Guardrails	Hard-coded safety rules in a healthcare AI system that override AI-generated content when patient safety is at risk. Non-negotiable boundaries that define what the AI must never do and when it must escalate to a human.
LLM (Large Language Model)	A type of AI model trained on large amounts of text data that can understand and generate human language. Examples include Claude, GPT, and Llama.
RAG	Retrieval-Augmented Generation. A technique where the AI retrieves relevant information from a knowledge base before generating a response, improving accuracy and reducing hallucination.
Health Literacy	The degree to which an individual can obtain, process, and understand basic health information needed to make appropriate health decisions. Low health literacy is associated with worse health outcomes.
HCAHPS	Hospital Consumer Assessment of Healthcare Providers and Systems. A standardized patient experience survey required by CMS. Scores affect Medicare reimbursement.
Care Plan	A documented plan of care for a patient that includes goals, interventions, and scheduled activities. In FHIR, represented as a CarePlan resource.
Medication Adherence	The extent to which a patient takes medications as prescribed by their healthcare provider. Non-adherence is a major driver of preventable hospitalizations.
Escalation Protocol	A defined process for transferring a patient interaction from AI to human care when the conversation exceeds the AI's safe operating scope.
Hallucination (AI)	When an AI model generates information that appears plausible but is factually incorrect. In healthcare AI, hallucinated medications or conditions are a safety risk.
Synthea	An open-source synthetic patient data generator by MITRE that produces realistic FHIR R4 clinical data without using real patient information.
Propel SDLC	An AI-assisted software development lifecycle framework that accelerates code generation, testing, and documentation.